MECHANICAL ENGINEERING · TECHNICAL PORTFOLIO

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RELIABILITY · PREDICTIVE DIAGNOSTICS · MRO · INDUSTRIAL OPTIMIZATION · MECHATRONICS



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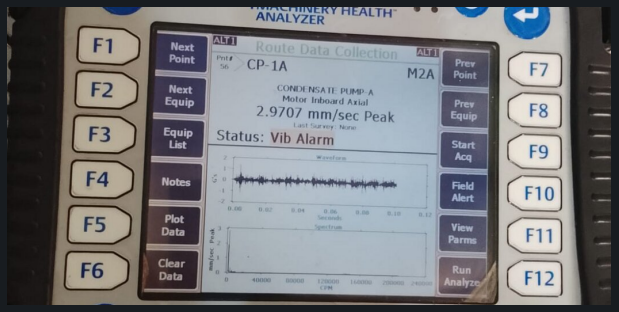
PROJECT 01 · PREDICTIVE DIAGNOSTICS · TURBO MACHINERY SERVICES INTERNSHIP

Vibration Diagnostics: Centrifugal Separator, Matco Foods

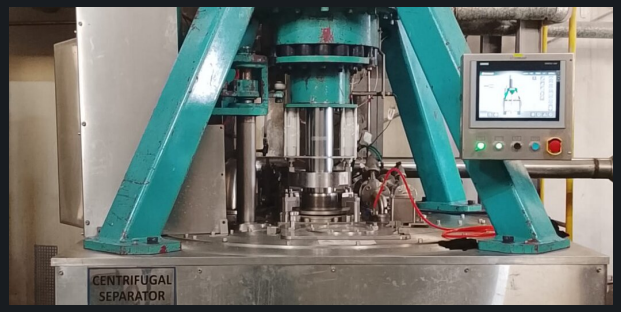
2.97 mm/s vs. 1.5 mm/s BASELINE

Field vibration diagnosis during a rotating-equipment internship at Turbo Machinery Services (Karachi), on a client job at Matco Foods Ltd.

MEASURED, CSI 2130 / DIGIVIBEMX



CSI 2130 route reading flags a Vib Alarm status at 2.9707 mm/s peak.



SUGROUP centrifugal separator, Matco Foods: asset under investigation.

MACHINE SYMPTOM

1.5
BASELINE, MM/S

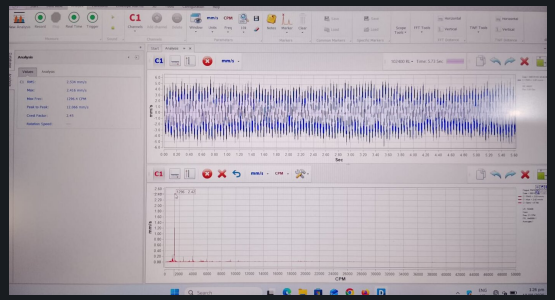
2.97
MEASURED PEAK, MM/S

A route check crossed the site's vibration-alarm threshold. The rest of the case works through one question: rotor imbalance, or something upstream of the rotor.

SIGNAL PROCESSING: TIME DOMAIN → FFT

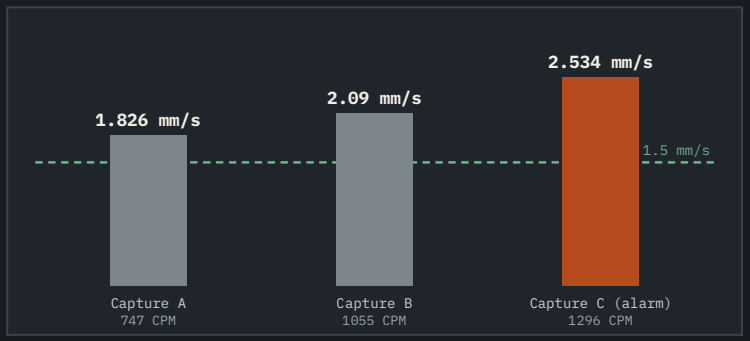


Time waveform: a slow, repeating amplitude envelope, not a steady, constant-amplitude signal.



FFT spectrum: one dominant low-order peak at 1296.4 CPM, RMS 2.534 mm/s.

READINGS ACROSS CAPTURES

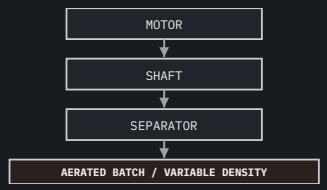


DIAGNOSTIC REASONING

Elevated vibration flagged on route → time waveform shows a modulated envelope rather than the steady 1X signature a clean rotor imbalance would produce → RMS and peak frequency shift across successive captures rather than holding constant → readings tracked against batch timing pointed to a faulty liquid glucose batch, not rotor imbalance, as the driver.

MECHANISM

CONCEPTUAL



Aerated batch → variable density → effective eccentricity → rotating force $F=m\omega^2 r$ → vibration, without a mechanical defect behind it.

ALSO ON THIS INTERNSHIP

ROTOR BALANCING - CEMB Z2000, ISO 1940
- Kaprico motor rotor, G6.3
- Hubco TDBF cartridge, G2.5 @ 5400 rpm

THERMAL SPRAY - Metco 16E / TAFA 400
- post sand-blast surface restoration

PROJECT 02 · MECHATRONICS · TEAM DESIGN CHALLENGE

Autonomous Pipe-Climbing Robot

1ST PLACE · AKDC CHALLENGE 2026

Abul Kalam Azad Design Challenge (IMechE), 2026, a five-member team design-build competition. Team Ascendants: M. Hassaan, Emmanuel Shamim, Ahsan Akram, Umair, M. Yasir. **TEAM PROJECT**

ENGINEERING QUESTION

How can a compact, four-wheel mechanism autonomously climb a 22 mm vertical pipe while holding sufficient contact force, and execute a controlled payload release, within a 400 mm × 300 mm envelope?

CONSTRAINTS

- Max envelope 400 mm dia. × 300 mm length
- Climb 22 mm vertical copper pipe
- Fully autonomous operation required
- Green LED continuously on during run
- Red LED + buzzer at defined events
- Friction force (μN) must exceed robot weight

FORCE CHAIN



Design constraint: Friction Force (μN) ≥ Weight of Robot

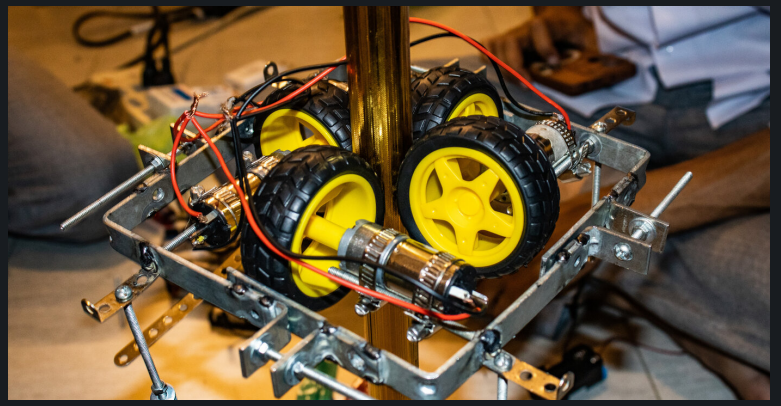
CONTROL LOGIC



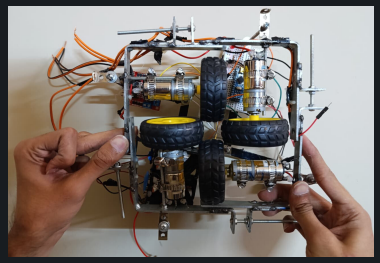
Sequence: climb → hold 10s at top (red LED + buzzer) → descend to midpoint → release chain → return to start

MECHANICAL DESIGN

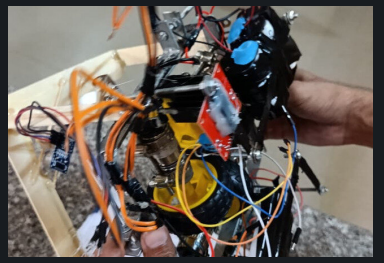
Rigid slotted-metal frame with movable brackets: tightening or loosening the mounting screws expands or contracts the wheel assembly, so the same frame adapts its contact force to different pipe diameters. Four motor-driven, rubber-tired wheels (67 mm dia.) press against the pipe in a symmetrical arrangement, keeping weight distribution balanced and resisting tilt while climbing. High-torque, low-speed gear motors provide the controlled vertical motion the friction budget requires.



Frame under test on a vertical pipe stand: four-wheel contact and adjustable bracket assembly.



Gear-motor and wheel assembly, frame build.



Top view: ultrasonic sensor mounts & hook/chain release.

SPEC SUMMARY

CONTROLLER	Arduino Nano
ACTUATION	4× gear motor + release servo
SENSING	Ultrasonic + limit switches
POWER	12V rechargeable battery
PAYLOAD	Hook-and-chain, servo release

CHALLENGES & SOLUTIONS

Slipping on the pipe
Rubber wheels + tuned normal force

Maintaining balance
Symmetrical mass distribution

Accurate release
Servo-controlled, fixed program position

Power efficiency
Motor/battery sized to torque budget

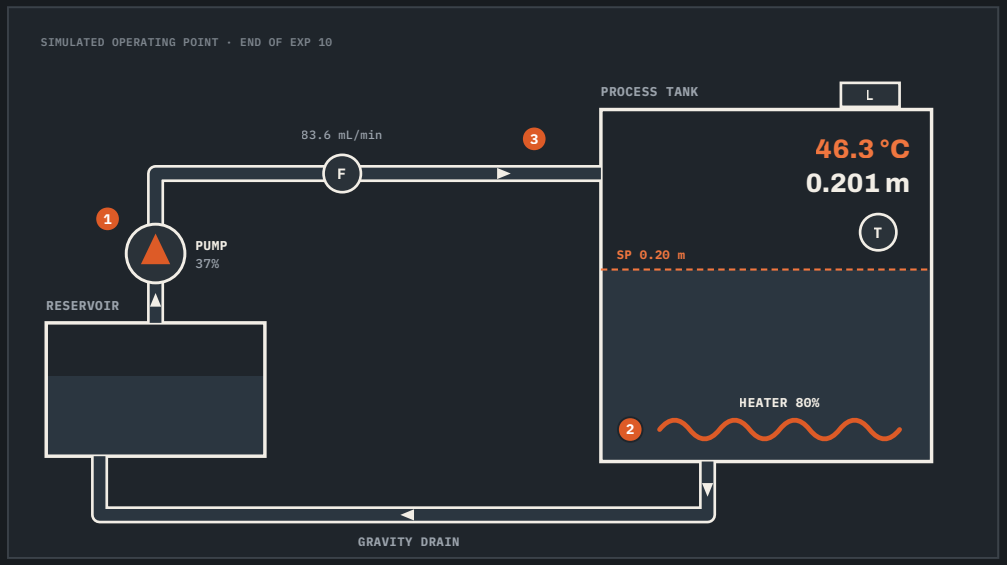
PROJECT 03 · PROCESS CONTROL & INSTRUMENTATION · INDEPENDENT SOFTWARE PROJECT

Aqualoop: Virtual Multi-Loop Process Control & Instrumentation

2 COUPLED LOOPS · 11 EXPERIMENTS · 62 TESTS

A Python process tank with a level PI loop and a temperature PID loop, imperfect virtual sensors, fault injection and a Streamlit dashboard.
Source: github.com/umairamir1305/aqualoop. SIMULATED DATA, NOT MEASURED

PROCESS SCHEMATIC · TWO LOOPS, ONE TANK

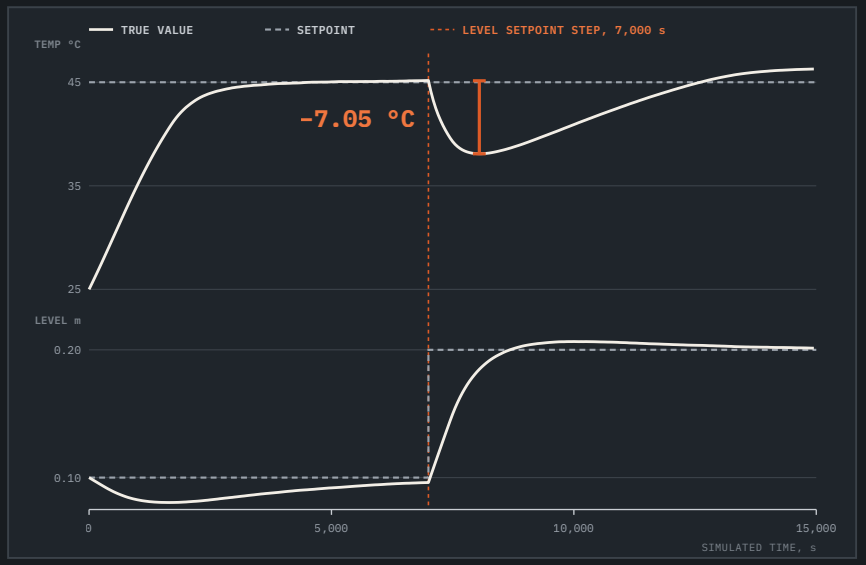


1 PUMP, DRIVEN BY LEVEL PI

2 HEATER, DRIVEN BY TEMPERATURE PID

3 INFLOW BRINGS COOLER WATER, SO THE LOOPS INTERACT

COUPLED RESPONSE · LEVEL STEP DISTURBS TEMPERATURE



RESULT · ALL VALUES SIMULATED

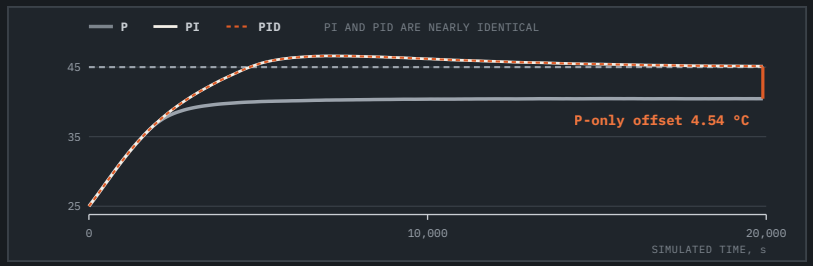
4.54 → 0.13 °C
STEADY-STATE ERROR, P-ONLY → PI / PID

8.0 %
TEMPERATURE OVERSHOOT, 25 → 45 °C STEP

5.1 °C
DIP, −15 °C INLET STEP; RECOVERS 2,220 s

7.05 °C
DIP CAUSED BY A 0.10 → 0.20 m LEVEL STEP

P VS PI VS PID · TEMPERATURE STEP



GOVERNING EQUATIONS

LEVEL · MASS BALANCE

$A \, dh/dt = Q_{in} - Cd \cdot Ao \cdot \sqrt{(2gh)}$

TEMPERATURE · ENERGY BALANCE

$m \cdot cp \cdot dT/dt = \rho \cdot Q_{in} \cdot cp \cdot (T_{in} - T) + Q_{heater} - Q_{loss}$

Explicit Euler · dt 1 s · seed 42

SCOPE

Simulation only. Plant parameters are engineering assumptions, not fitted to a physical rig.

Reproducible. 62 tests pass and all 11 experiments regenerate from one command.

PROJECT 04 · RELIABILITY ENGINEERING · PIA ENGINEERING & MAINTENANCE INTERNSHIP

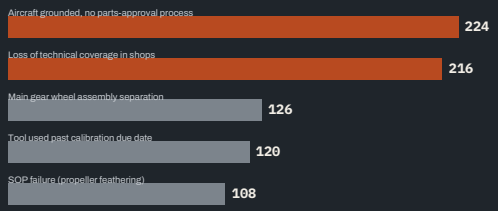
Aircraft MRO Reliability Engineering: FMEA, RCA & RCM/MSG-3

16 FAILURE MODES · 3 METHODS · 1 REAL INCIDENT

Three independent case studies built from a four-week internship rotation through Base Maintenance, Components Maintenance Division, and Powerplant Overhaul. Ratings and conclusions are the author's own application of each method to field observation, not PIA's internal reliability data. **INTERNSHIP: PIA ENGINEERING & MAINTENANCE**

FACILITY-WIDE FMEA

16 failure modes across five functional areas, scored on Severity × Occurrence × Detection (1-10 each).



The two highest-RPN entries are organizational gaps with **no formal control at all**; both outrank the most severe mechanical failure mode on the table.

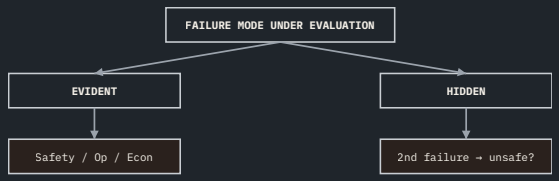
ROOT CAUSE ANALYSIS METHODOLOGY

5-Why, Fishbone, and Fault Tree Analysis run side by side on the same recurring problem: a bench-test component missing its 3-day turnaround target (slipping to 7-10 days).

5-WHY	Fast	1 chain
FISHBONE	Broad	No ranking
FAULT TREE	Slow	Logic-exact

All three converge on the same fixable cause: the fault coupon has no field to flag a component as urgent, and no process revisits documentation when the workflow changes.

RCM & MSG-3 DECISION LOGIC



Task order: Lubrication/Servicing → On-Condition → Hard-Time → Condition-Monitoring

ENGINE/APU: on-condition (chip detectors + borescope)
BRAKES: hybrid (700-800 landings + wear-pin + thermal sensor)
WHEELS/TIRES: hard-time (2,500 landings + multi-method NDT)

ONE INCIDENT, THREE LENSES

PK-306

A320 · Karachi-Lahore · Mar 2025

A PIA A320 landed safely in Lahore after departing Karachi missing a main-gear wheel; no injuries. The event runs through all three studies: FMEA maps it to the highest-severity mechanical entry on the table (wheel assembly separation); RCA compares it against a hypothetical delay scenario as a worked real-world example of causal-method rigor; RCM/MSG-3 uses it to show why hard-time retirement, multi-method NDT, and engineered fail-safes (fuse plug, redundant gear geometry) exist as layered controls rather than one line of defense. The public root cause was never conclusively confirmed (reported contributing factors span foreign-object debris, a possible material or manufacturing defect, and maintenance-related factors), and that uncertainty is treated as data, not glossed over.

BATHTUB CURVE: WHY EACH TASK TYPE FITS WHERE IT DOES



Engine/APU failure modes sit in the flat middle (on-condition wins); wheels, tires and life-limited rotor parts sit in the rising tail (hard-time wins); engineered fail-safes cover the early-life gap no schedule can close.

MATERIAL & RELIABILITY TRADE-OFF

BRAKE DISCS - steel: cheaper, shorter life (ATR 42)
 - carbon: lighter, higher heat capacity (ATR 72+)
TIRES - Bridgestone: 300-350 landings
 - Michelin: 500-550 landings
 - both retreadable up to 3×

A longer-life material only pays for itself if the maintenance program is built to extract that life, not retired conservatively out of habit.

DISPATCH URGENCY & MEASUREMENT TRUST

MEL CAT A - fix within 24 hr
MEL CAT B - fix within 3 days
MEL CAT C - fix within 10 days
MEL CAT D - fix within 120 days

Same severity-to-urgency logic as MSG-3, applied to dispatch. Every on-condition task above is only as good as the tool measuring it; handheld gear is recalibrated every 6 months, fixed tooling every year.

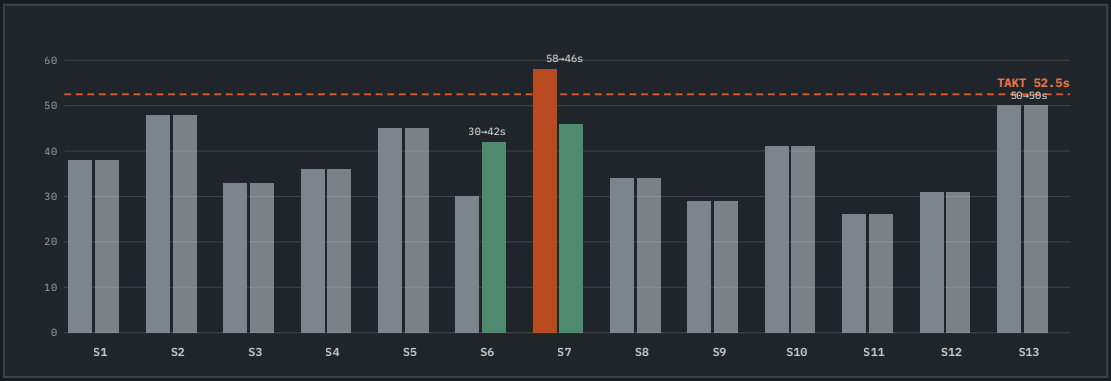
PROJECT 05 · INDUSTRIAL OPTIMIZATION · INDEPENDENT CASE STUDY

Value Stream Mapping, Takt Time & Line Balancing

66% → 77% BALANCING EFFICIENCY

ILLUSTRATIVE DATA Independent methodology case study set in a motorcycle final-assembly context (Atlas Honda, Pakistan). Station cycle times, takt time, buffer sizes, and lead time are constructed by the author to demonstrate the method, not confidential production records or a claim about any company's actual current-state line.

BOTTLENECK REBALANCE: 13-STATION FINAL ASSEMBLY VS. TAKT TIME



Station 7 (electrical harness) ran at 58s, 110% of the 52.5s takt time, and paced the whole line. Moving roughly 12s of preliminary harness-routing work to Station 6 (handlebar/controls, which had 22.5s of spare capacity) respects assembly precedence and uses no new equipment. The new bottleneck, Station 13, sits at 95% of takt.

METHOD CHAIN

VSM: where is time lost across the whole flow?

Takt Time: what rhythm does demand require? (52.5 s)

Line Balancing: how should work be spread to hold it?

RESULT

66.2 → 76.8%
BALANCING EFFICIENCY

255 → 151s
IDLE TIME / UNIT

6.4 → 3.5hr
MODELLED LEAD TIME

2.2 → 4.0%
VALUE-ADDED RATIO

PRECEDENCE-CONSTRAINED MOVE

S6 (handlebar/controls, 30s) immediately precedes S7 and had 22.5s of spare capacity against takt. Preliminary harness routing along the handlebar area needs the handlebar in place, not the ignition or lighting circuits connected, so it can move upstream to S6 without violating assembly order. S6: 30→42s. S7: 58→46s. Total work content is unchanged at 499s across all 13 stations; only the two adjusted stations move.

WHERE THE LEAD-TIME GAIN ACTUALLY COMES FROM

The rebalance fixes takt compliance, not lead time on its own; total processing stays at 499s. The ~46% lead-time cut is driven mainly by shrinking two of the four buffers with kanban-style pull signals, since buffer stock, not assembly work, made up the large majority of the original 6.4-hour lead time.

WHERE THE BUFFERS ACTUALLY MOVED

BUFFER 1 · Frame Store	BUFFER 2 · Sub-Assembly	BUFFER 3 · Pre-Paint	BUFFER 4 · Finished Goods
0.30 → 0.15 days	**0.20 → 0.10 days**	**0.15 → 0.08 days**	**0.25 → 0.15 days**

INTERVIEW DEFENSIBILITY: WHAT THIS CASE STUDY DOES AND DOESN'T CLAIM

- Demonstrates the VSM → takt → line-balancing method end to end on a representative 13-station line
- Every number traces to a stated assumption or a simple calculation, not a black box
- Does not claim access to any company's actual production data or floor time
- Does not claim the line-balancing move alone produced the lead-time result; buffer policy did most of that work